

NexusFinance — Platform Report

Digital Lending Platform for Cambodia

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Full Project Report • Features • Architecture • Integrations

Table of Contents

1. Project Overview
2. Portal Features
3. Authentication & User Management
4. Payment Integration
5. Telegram Bot Features
6. Mobile Application
7. Hosting & Infrastructure

1. Project Overview

NexusFinance is a comprehensive fintech lending platform designed for microfinance institutions in Cambodia. The platform serves three user roles — Customers, Loan Officers, and Super Administrators — each with a dedicated portal tailored to their workflow.

2. Portal Features

2.1 Customer Portal

- Loan application with step-by-step form (personal info, financial details, loan terms)
- Outstanding balance tracking and wallet management
- KHQR payment integration (Bakong-compatible)
- Transaction history and repayment scheduling
- Profile management (name, email, phone, password)

2.2 Loan Officer Portal

- Application review dashboard with priority filtering
- Credit score analysis and DTI ratio calculation
- Direct loan approval/rejection/hold workflow
- KYC task management

2.3 Super Admin Portal

- Platform-wide analytics and volume tracking
- Interest rate and auto-underwrite configuration
- User management (roles, password resets)
- Full audit log with timestamped entries
- System configuration

3. Authentication & User Management

Feature	Implementation
Registration	Email + password → Brevo OTP verification (10-min expiry)
Login	Email + password with bcrypt comparison, JWT issued — blocked until email is verified
Email Verification	New accounts cannot log in until the emailed OTP is verified

Password Reset	Brevo OTP → verify code → set new password
Profile Update	PATCH /api/auth/profile (name, email, phone)
Password Change	OTP verification required before password update
Role System	customer, loan-officer, super-admin with server-side enforcement

4. Payment Integration

4.1 KHQR (Bakong)

KHQR is Cambodia's national QR payment standard based on EMVCo. NexusFinance generates KHQR codes for loan repayments and can verify payments via the Bakong API.

- Static and dynamic QR generation
- Payment verification via MD5 hash
- Deeplink generation for mobile banking apps

4.2 ABA PayWay

ABA Bank's PayWay Purchase API powers hosted checkout for loan repayments. The customer selects an amount and is redirected to ABA's secure hosted checkout page (view_type: hosted_view), where they can pay via KHQR, ABA PAY, or cards.

- Sandbox mode currently active (checkout-sandbox.payway.com.kh)
- Hosted checkout with view_type=hosted_view and payment_gate=0
- HMAC-SHA512 signature on every purchase request (24-field fixed hash order)
- Return / cancel redirects to /payment/success and /payment/cancel
- Webhook callback + GET return handler both verify and persist payments
- Approved payments are persisted to the ledger (nexus_transactions) and trigger Telegram confirmation

5. Telegram Bot Features

Command	Access	Description
/start	All users	Welcome message and quick actions
/link	All users	Link Telegram account — emails a code to your NexusFinance profile

/confirm <code>	All users	Complete linking with the emailed code
/unlink	All users	Unlink Telegram account
/status	All users	View linked account status
/reminder-check	Admin	Run payment reminder check now
/broadcast	Admin	Send message to all linked users
/send	Admin	Send message to a specific user
/stats	Admin	Platform statistics
/loans	Admin	Recent loan applications
/users	Admin	Registered users list
/notifications	Admin	Recent payment notifications
/help	All users	Show available commands

6. Mobile Application

The Android app is an Expo WebView shell that loads the live web application. This allows the platform to be distributed as an installable APK while keeping all logic on the web server.

- Expo SDK 57 with React Native WebView
- Full-screen immersive design matching the web app
- Offline retry screen with brand styling
- Custom app icon and splash screen

7. Hosting & Infrastructure

Component	Service	Details
Frontend	Vercel	React SPA, auto-deploys from GitHub main branch
Backend	Render	Node.js/Express, free tier with UptimeRobot keep-alive
Database	Supabase	PostgreSQL with Row Level Security
Email	Brevo	Transactional OTP emails, 300/day free tier
Monitoring	UptimeRobot	5-minute health checks on /api/health

Bot	Telegram API	node-telegram-bot-api with long polling
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